

Equilibria, Stability, and Discrete-Time Dynamics

Optimization and Learning for Robot Control

RCI Lab

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Previously:

$$\dot{x} = f(x, u)$$

Manipulator form:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = B\tau$$

Linear systems:

$$\dot{x} = Ax + Bu$$

Definition:

An equilibrium point (x_e, u_e) satisfies

$$f(x_e, u_e) = 0$$

Then:

$$\dot{x} = 0$$

System remains at rest.

Example: Pendulum Equilibria

Pendulum dynamics:

$$m\ell^2\ddot{\theta} + mg\ell \sin \theta = \tau$$

At equilibrium:

$$\ddot{\theta} = 0$$

Thus:

$$mg\ell \sin \theta_e = \tau_e$$

For $\tau_e = 0$:

$$\sin \theta_e = 0 \Rightarrow \theta_e = 0, \pi$$

Multiple equilibria.

First Control Question

Can we move the equilibrium?

Solve:

$$f(x_e, u_e) = 0$$

This is a nonlinear root-finding problem.

Control can shift equilibria.

Stability of Equilibria

Question:

If $x(0)$ is near x_e , does $x(t)$ stay near x_e ?

Linearize:

$$\dot{x} = f(x)$$

$$\delta \dot{x} = \left. \frac{\partial f}{\partial x} \right|_{x_e} \delta x$$

Define Jacobian:

$$A = \frac{\partial f}{\partial x}$$

Linear Stability Condition

Linearized system:

$$\delta\dot{x} = A\delta x$$

Solution:

$$\delta x(t) = e^{At} \delta x(0)$$

Stability condition:

$$\operatorname{Re}(\lambda_i(A)) < 0$$

All eigenvalues must have negative real part.

Pendulum Stability

Linearize around $\theta = 0$:

$$\sin \theta \approx \theta$$

Then:

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0$$

Eigenvalues:

$$\lambda = \pm i\sqrt{\frac{g}{\ell}}$$

Purely imaginary $\rightarrow$ undamped oscillation (marginally stable).

Adding Damping

Add damping torque:

$$\tau = -k_d \dot{\theta}$$

Then:

$$\ddot{\theta} + k_d \dot{\theta} + \frac{g}{\ell} \theta = 0$$

Eigenvalues:

$$\lambda = \frac{-k_d \pm \sqrt{k_d^2 - 4g/\ell}}{2}$$

If $k_d > 0$:

$$\operatorname{Re}(\lambda) < 0 \Rightarrow \text{stable}$$

In simulation:

$$\dot{x} = f(x)$$

We approximate using timestep h .

Forward Euler:

$$x_{k+1} = x_k + hf(x_k)$$

Discrete-Time Linearization

Continuous:

$$\dot{x} = Ax$$

Forward Euler:

$$x_{k+1} = x_k + hAx_k$$

$$x_{k+1} = (I + hA)x_k$$

Define:

$$A_d = I + hA$$

Discrete Stability Condition

Discrete system:

$$x_{k+1} = A_d x_k$$

Stability requires:

$$\rho(A_d) < 1$$

where ρ is spectral radius.

Thus:

$$|\lambda_i(A_d)| < 1$$

Eigenvalues must lie inside unit circle.

Why Forward Euler Can Blow Up

Continuous eigenvalue:

$$\lambda = i\omega$$

Discrete eigenvalue:

$$\lambda_d = 1 + hi\omega$$

Magnitude:

$$|\lambda_d| = \sqrt{1 + h^2\omega^2} > 1$$

Therefore:

Forward Euler destabilizes oscillatory systems.

Runge-Kutta 4 (RK4)

Higher-order integrator:

$$k_1 = f(x_k)$$

$$k_2 = f\left(x_k + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(x_k + \frac{h}{2}k_2\right)$$

$$k_4 = f(x_k + hk_3)$$

Update:

$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Takeaways

- Stability depends on eigenvalues.
- Continuous stability: $\text{Re}(\lambda) \leq 0$.
- Discrete stability: $|\lambda| < 1$.
- Forward Euler can destabilize systems.
- Always sanity-check energy and behavior.